Customer Purchase Behavior Analysis using Excel Dashboard

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **"Customer Purchase Behavior Analysis using Excel Dashboard"** is the bonafide work of **Partistha Singhmar**, student of Bachelor of Computer Applications, Chandigarh University, who carried out the project under supervision.

(Signature)
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List of Standards (Mandatory For Engineering Programs)

Standard	Publishing Agency	About the standard	Page no
IEEE 802.11	IEEE	IEEE 802.11 is part of the IEEE 802 set of local area network (LAN) technical standards and specifies the set of media access control (MAC) and physical layer (PHY) protocols for implementing wireless local area network (WLAN) computer communication.	Mention page nowhere standard is used

Note: Text in Red is presented as an example (replace with relevant information)

ABSTRACT

In the modern era of digital transformation, data plays a vital role in decision-making processes across industries. This project focuses on analyzing customer purchase behavior using Microsoft Excel tools such as Pivot Tables, Charts, and Interactive Dashboards.

The dataset used in this project contains demographic and behavioral information about customers, including age, gender, marital status, income, education, occupation, commute distance, region, and purchasing decisions. The main objective of this project is to transform raw data into meaningful insights through data cleaning, preprocessing, and visualization.

The project follows a structured approach, beginning with preserving the original dataset, followed by creating a working sheet for data cleaning. Various transformations such as removing duplicates, standardizing categorical values, formatting numerical data, and generating new attributes like age brackets were performed.

Further, Pivot Tables were created to analyze different aspects of customer behavior such as income trends, commute patterns, and age distribution. These analyses were then visualized using charts and integrated into an interactive dashboard. Slicers were implemented to allow dynamic filtering based on marital status, region, and education level.

The results demonstrate that Excel can effectively be used for data analysis and visualization, enabling users to derive actionable insights. This project highlights the importance of data preprocessing and visualization in understanding customer behavior and supporting business decisions.

CHAPTER 1.

INTRODUCTION

1.1 Identification of Client / Need / Relevant Issue

In today's competitive business environment, understanding customer behavior has become essential for organizations. Businesses rely heavily on data to understand customer preferences, purchasing patterns, and demographic trends.

The dataset used in this project represents customer information related to purchasing decisions. This type of analysis is highly relevant in industries such as retail, e-commerce, and marketing, where companies aim to optimize their strategies based on customer insights.

The need for this project arises from the challenge of converting raw, unstructured data into meaningful information. Without proper analysis, data remains unused and does not contribute to decision-making.

1.2 Identification of Problem

The major problem addressed in this project is the difficulty in analyzing raw customer data and extracting meaningful insights. Raw datasets often contain inconsistencies, duplicate entries, and unstructured values, making analysis complex.

Specific problems include:

- Presence of duplicate records
- Inconsistent categorical values (e.g., M, F, S)
- Difficulty in interpreting numerical data
- Complexity in analyzing raw age values
- Lack of visualization for better understanding

1.3 Identification of Tasks

The project involves the following tasks:

- Creating structured worksheets for data handling
- Cleaning and preprocessing the dataset
- Standardizing categorical variables
- Creating new calculated columns (Age Brackets)
- Performing data analysis using Pivot Tables
- Designing charts for visualization
- Developing an interactive dashboard
- Implementing slicers for dynamic filtering

CHAPTER 1.

INTRODUCTION

1.4 Timeline

The project was completed in multiple stages:

1. Understanding the dataset
2. Creating working sheets
3. Data cleaning and preprocessing
4. Feature engineering
5. Data analysis using Pivot Tables
6. Dashboard creation
7. Report preparation

Each phase was completed sequentially to ensure accuracy and clarity in results.

1.5 Organization of the Report

This report is organized into five chapters:

- Chapter 1: Introduction and problem definition
- Chapter 2: Literature review and background study
- Chapter 3: Design flow and methodology
- Chapter 4: Results and analysis
- Chapter 5: Conclusion and future work

CHAPTER 2.

LITERATURE REVIEW/BACKGROUND STUDY

2.1 Background of Data Analysis

Data analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information. It plays a significant role in decision-making across various industries.

Microsoft Excel is one of the most widely used tools for data analysis due to its accessibility and powerful features.

2.2 Existing Solutions

Several tools are available for data analysis:

- Microsoft Excel
- Power BI
- Tableau
- Python (Pandas, NumPy)

Among these, Excel is preferred for small to medium datasets due to its simplicity.

2.3 Limitations of Existing Solutions

While Excel is powerful, it has limitations:

- Limited handling of large datasets
- Less automation compared to programming tools
- Manual effort required for data cleaning

2.4 Review Summary

The literature suggests that Excel remains an effective tool for beginners and intermediate users for performing data analysis and visualization.

2.5 Problem Definition

To analyze customer purchase data using Excel and develop an interactive dashboard that provides insights into customer demographics and behavior.

CHAPTER 2.

LITERATURE REVIEW/BACKGROUND STUDY

2.6 Goals / Objectives

- To clean and preprocess raw data
- To create meaningful categories for analysis
- To analyze customer purchasing trends
- To design an interactive dashboard

CHAPTER 3.

DESIGN FLOW/PROCESS

3.1 Dataset Structure and Sheet Creation

Four sheets were created:

1. **Car Buyers Sheet** – Original dataset (unchanged)
2. **Working Sheet** – Data cleaning and processing
3. **Pivot Table Sheet** – Data analysis
4. **Dashboard Sheet** – Visualization

3.2 Data Cleaning Process

Data cleaning involved:

- Removing duplicate records
- Replacing categorical values:
  - M → Male
  - F → Female
  - S → Single
- Formatting income as currency
- Removing decimal points for better readability

3.3 Feature Engineering

A new column **Age Brackets** was created using IF function:

- Adolescence
- Middle Age
- Old Age

This simplified analysis and improved visualization.

3.4 Pivot Table Creation

Multiple Pivot Tables were created to analyze:

- Average income of purchasers
- Customer commute distance
- Customer age brackets
- Additional comparative analysis

CHAPTER 3.

DESIGN FLOW/PROCESS

3.5 Chart Design

Charts were created for each Pivot Table, including:

- Bar charts
- Pie charts
- Column charts

These charts helped in visualizing patterns effectively.

3.6 Dashboard Development

A dashboard was created by combining all charts.
A card-based layout was used to improve readability.

3.7 Slicers and Interactivity

Slicers were added for:

- Marital Status
- Region
- Education

Using report connections, slicers control all charts simultaneously.

CHAPTER 4.

RESULTS ANALYSIS AND VALIDATION

4.1 Income Analysis

The analysis shows that customers with higher income levels tend to have higher purchasing behavior. This indicates a direct relationship between income and purchasing decisions.

4.2 Commute Distance Analysis

Customers with shorter commute distances show different purchasing patterns compared to those with longer commute distances. This may indicate lifestyle differences.

4.3 Age Bracket Analysis

The use of age brackets simplified the analysis:

- Middle-aged individuals contribute significantly to purchases
- Younger individuals show moderate purchasing
- Older individuals show lower purchasing activity

4.4 Dashboard Insights

The dashboard provides:

- Interactive filtering
- Real-time updates
- Easy comparison

Users can analyze data dynamically using slicers.

4.5 Validation

The results were validated by cross-checking Pivot Tables and ensuring consistency in data.

CHAPTER 5.

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project demonstrates how Excel can be used effectively for:

- Data cleaning
- Data analysis
- Visualization

The dashboard provides meaningful insights into customer behavior.

5.2 Future Work

- Use Power BI for advanced visualization
- Apply machine learning techniques
- Automate data processing

  - Automation

    Reports can be automated using macros or scripts.

  - Integration with Database

    Data can be connected to real-time databases for dynamic updates.

REFERENCES

The following references were used during the development and understanding of concepts related to this project:

1. **Microsoft Corporation.**
   Microsoft Excel Documentation and Support.
   Available at: https://support.microsoft.com/excel
2. **Walkenbach, John.**
   Microsoft Excel 2019 Bible. Wiley Publications.
3. **Alexander, Michael & Kusleika, Dick.**
   Excel 2019 Power Programming with VBA. Wiley.
4. **Jelen, Bill & Syrstad, Tracy.**
   Excel Pivot Tables and Pivot Charts. Pearson Education.
5. **Online Learning Platforms:**
   - YouTube tutorials on Excel Dashboard Creation
   - Coursera – Data Analysis Fundamentals
   - Udemy – Excel for Data Analysis
6. **Data Analysis Concepts:**
   - Kotu, Vijay & Deshpande, Bala.
     Predictive Analytics and Data Mining. Morgan Kaufmann
7. **General Data Visualization Principles:**
   - Few, Stephen.
     Information Dashboard Design. Analytics

APPENDIX

Appendix A: Dataset Description

The dataset used in this project contains customer-related information, including:

- Customer ID
- Gender
- Marital Status
- Income
- Children
- Education
- Occupation
- Home Ownership
- Number of Cars
- Commute Distance
- Region
- Age
- Purchase Decision

This dataset was used to analyze customer purchasing behavior.

Appendix B: Data Cleaning Steps

The following preprocessing steps were applied:

- Removal of duplicate entries
- Standardization of categorical values:
  - M → Male
  - F → Female
  - S → Single
- Formatting of income values into currency
- Removal of decimal values for simplicity
- Creation of a new column “Age Brackets”

Appendix C: Pivot Tables Created

The following Pivot Tables were used in the project:

1. Average Income of Purchasers
2. Customer Commute Distance Analysis
3. Customer Age Bracket Distribution
4. Comparative Analysis based on Demographics

Appendix D: Dashboard Components

The dashboard includes:

- Multiple charts (bar, pie, column charts)
- Slicers for filtering:
  - Marital Status
  - Region
  - Education
- Interactive visualization connected through report connections

Appendix E: Tools Used

- Microsoft Excel
- Pivot Tables
- Charts and Graphs
- Slicers
- Data Cleaning Techniques

USER MANUAL

(Complete step by step instructions along with pictures necessary to run the project)

1. Introduction

This user manual provides step-by-step instructions to understand and use the Excel-based dashboard developed in this project. The dashboard allows users to analyze customer data interactively.

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2. System Requirements

- Microsoft Excel (2016 or later recommended)
- Basic knowledge of Excel

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3. File Structure

The Excel file contains the following sheets:

1. **Car Buyers Sheet**
    - Contains original dataset
    - No modifications made
2. **Working Sheet**
    - Contains cleaned and processed data
3. **Pivot Table Sheet**
    - Contains analysis using Pivot Tables
4. **Dashboard Sheet**
    - Contains final visualizations

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4. How to Use the Dashboard

Step 1: Open the Excel File

- Open the project Excel file in Microsoft Excel

Step 2: Navigate to Dashboard Sheet

- Click on the “Dashboard” tab at the bottom

USER MANUAL

Step 3: Use Slicers for Filtering

You will see slicers on the dashboard:

- **Marital Status** (Married / Single)
- **Region** (Europe, North America, Pacific)
- **Education** (Bachelor, Graduate Degree, High School, Partial College, Partial High School)

 Click on any option to filter data
 Hold Ctrl to select multiple options

Step 4: Observe Chart Changes

- All charts update automatically based on slicer selection
- This helps in comparing different customer segments

Step 5: Reset Filters

- Click the filter clear icon in slicers to reset

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5. Understanding the Charts

- **Income Chart:** Shows average income of customers
- **Commute Chart:** Displays distance traveled by customers
- **Age Bracket Chart:** Shows distribution of age groups

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6. Working with Pivot Tables

- Navigate to Pivot Table sheet
- You can modify filters or fields
- Drag and drop fields to customize analysis

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7. Limitations

- Works best with structured data
- Large datasets may slow performance
- Requires manual updating if data changes

USER MANUAL

8. Troubleshooting

- If charts are not updating → Check slicer connection
- If data is missing → Refresh Pivot Tables
- If formatting is incorrect → Reapply formatting

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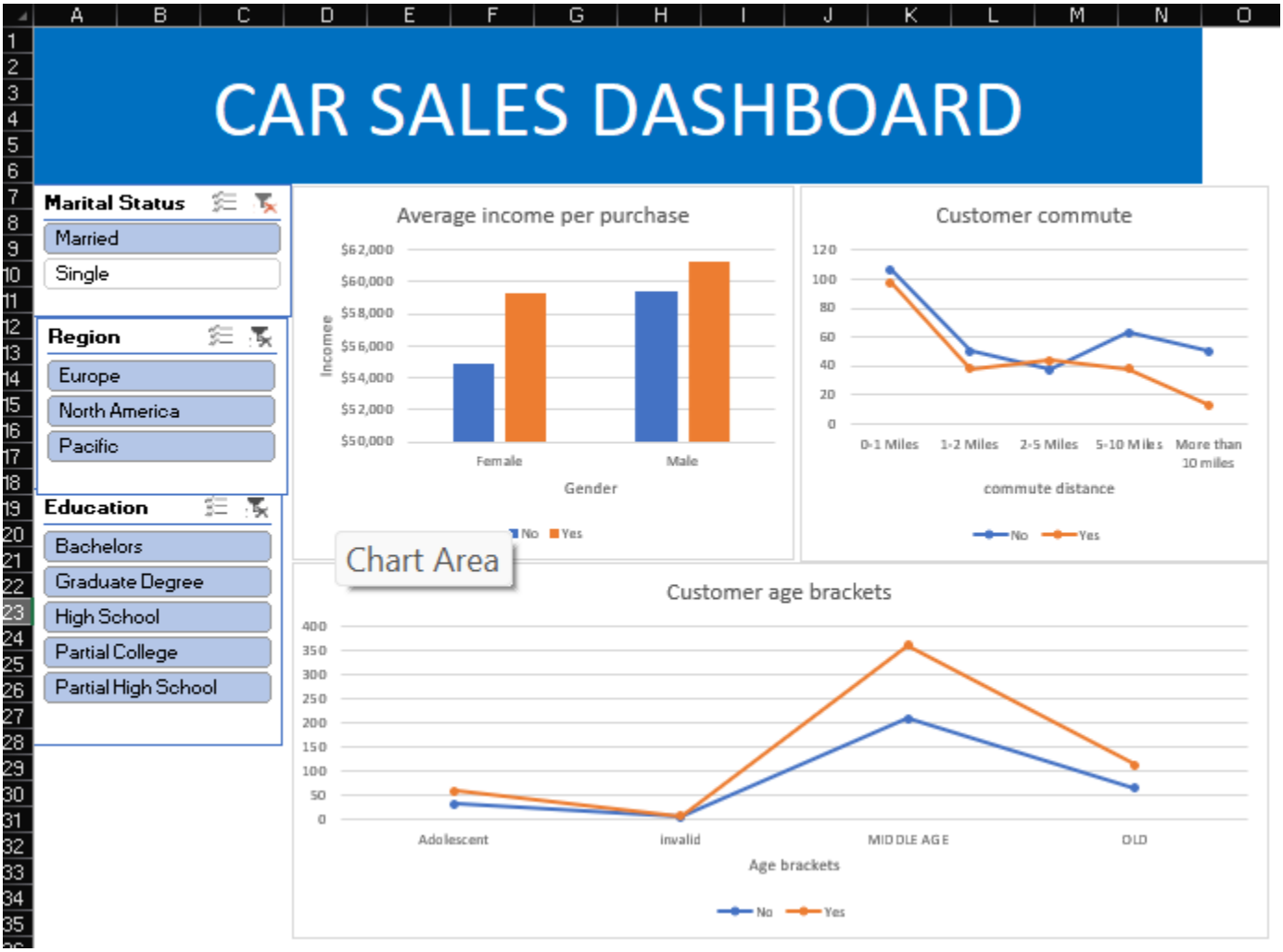
9. Conclusion

The dashboard provides a simple and interactive way to analyze customer data using Excel. It helps users gain insights without requiring advanced technical knowledge

- Save the file in .xlsx format

USER MANUAL

IMAGES